INTERNATIONAL MONETARY FUND

PAPUA NEW GUINEA

May 25, 2022

STAFF REPORT FOR THE 2022 ARTICLE IV CONSULTATION AND REVIEW OF THE STAFF MONITORED PROGRAM—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staff of the International Monetary Fund (IMF) and the International Development Association (IDA)[1,2]

Papua New Guinea Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgment	*No*

Papua New Guinea (PNG) remains at high risk of debt distress under the Low-Income Country Debt Sustainability Framework (LIC DSF), with weak debt-carrying capacity. While the planned fiscal consolidation helps address debt vulnerabilities exacerbated by the global COVID-19 shock, the risk of both external and public debt distress continues to be assessed as high. Over the medium-term, public debt enters a downward trend and the projected temporary breaches of sustainability indicators can mostly be addressed by debt management operations as well as improvements in revenue generation. The Debt Sustainability Analysis (DSA) suggests that PNG is susceptible to exports and other shocks, signaling downside risks to the debt outlook in a global environment of high uncertainty. To lower the risk of debt distress and ensure debt sustainability, gradual fiscal consolidation, including boosting revenues, and steadfast structural reforms to promote private sector growth would be needed. Conditional on the implementation of the authorities plans for further fiscal consolidation and conservative financing strategies, PNG's external and overall debt is judged as sustainable.

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[1] The Composite Indicator (CI) of 2.62 is based on the latest available CI information—April 2022 IMF World Economic Outlook (WEO) and the World Bank's Country Policy and Institutional Assessment (CPIA) for 2021, indicating a "weak" capacity to carry debt.

[2] This Debt Sustainability Analysis has been prepared jointly by the International Monetary Fund and the World Bank, in accordance with the revised Debt Sustainability Framework for low-income countries approved by the Executive Boards of the IMF and the International Development Association.

PUBLIC SECTOR DEBT COVERAGE

1. The coverage of public debt in the DSA is unchanged from the previous (December 2021) DSA (Text Table 1). The segments of the public sector captured in the DSA include the central government, state and local government, and to some extent guarantees to other entities in the public and private sector, including parts of state-owned enterprises (SOEs). However, debt numbers do not fully capture implicit government guaranteed debts of SOEs and unfunded superannuation liabilities relating to pensions. For the purposes of this DSA, the coverage of public sector debt remains unchanged from the last DSA, which was prepared in December 2021 in the context of a request for an IMF Staff Monitored Program (SMP). Given continued difficulties in capturing and assessing SOE risks, a contingent liabilities stress test is included in this DSA, assuming 9 percent of GDP as SOE debt is not captured in official public debt data, and 3 percent of GDP for other elements of general government (mainly unfunded superannuation liabilities related to pensions). Separately, according to the World Bank's PPP database, the PPP capital stock in PNG is zero and, therefore, no default shock is triggered. A financial market shock of 5 percent is added, reflecting the average fiscal cost of financial crisis in low-income countries. With these assumptions, the cumulative shock in the contingent liabilities stress test amounts to 17 percent of GDP—compared to 7 percent under default assumptions.

Text Table 1. Papua New Guinea: Coverage of Public Debt

	Subsectors of the public sector	Subsection Covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central, state, and local governments, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	3.0	Unfunded superannuation liabilities, relating to pensions.
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	9.0	SOE sovereign guarantee in dispute.
4	PPP	35 percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		17.0	

BACKGROUND ON DEBT

2. Between 2017 and 2021, the stock of public debt in PNG increased from around 25 billion Kina to 49 billion Kina. This was mainly due to external loans, while the creditor composition has been gradually shifting away from commercial loans towards official multilateral and bilateral financing. Shocks from the pandemic, including the terms-of-trade shock to PNGs main exports (mainly LNG and metals)

have exacerbated fiscal deficits and increased government debt, resulting in a gross nominal public debt path that is higher than at the time of the 2019 Article IV consultation. However, expensive commercial loans have been replaced with official multilateral and bilateral financing at more favorable conditions. This has helped to improve debt stability indicators. PNG participated in the G-20 Debt Service Suspension Initiative (DSSI) which lowered debt service costs during 2020 and 2021. Public and Publicly Guaranteed (PPG) external debt figures used for this DSA are consistent with the information in the World Bank's International Debt Statistics.

3. The IMF's general allocation of Special Drawing Rights (SDRs) became effective in August 2021, with SDR 252 million (US$ 357 million, or 95.7 percent of quota) allocated to PNG. The authorities decided to use the full SDR allocation to support the 2021 budget. They used the SDR allocation to replace costly financing, while reducing the budget deficit. The SDR allocation is included in public debt numbers for the purpose of this DSA (international reserves with the central bank increase temporarily until the SDRs are converted into hard currency and used), and the associated debt service for the amount outstanding is reflected as well.

4. PNG is an IDA blend country, with total IDA19 allocation at SDR 178.4 million. IDA's Performance-Based Allocation (PBA) for PNG benefitted from resource frontloading amid a shortening of IDA19 to two years as well as an intra-regional reallocation for SDR 0.6 million, approved in November 2021. The IDA decision to graduate a country to IBRD-only status is based on an assessment of the country's macroeconomic prospects, risk of debt distress, vulnerability to shocks, institutional constraints, and levels of poverty and social indicators.

5. The IDA Sustainable Development Finance Policy (SDFP) supports PNG in addressing key debt vulnerabilities. This may allow future access to full IBRD borrowing. As part of the SDFP, PNG has satisfactorily implemented the Performance and Policy Actions (PPA) for fiscal year 2021. These were aimed at improving debt management and debt transparency. PNG has two further SDFP PPAs for fiscal year 2022: (i) a US$ 1 billion non-concessional PPG borrowing limit for new non-concessional long-term contractual obligations, which applies continuously throughout FY22, and (ii) to operationalize the 2021 State Guarantee Policy (supported by the SDPF in FY21) with the adoption of a binding Guidance Note for Loan Guarantees. The Guidance Note would be the most important step for functionality of the Guarantee Policy, with the overall objective to incorporate credit risk assessment in decision making on guarantees and to improve reporting by guaranteed entities.

BACKGROUND ON MACROECONOMIC FORECASTS

6. Economic activity in 2020 was significantly disrupted by the pandemic but has rebounded since then (*Text Table 2*). Real GDP in 2021 is estimated to have increased by 1.2 percent due to stronger demand for PNG's key exports as well as high global commodity prices. The pre-crisis forecast (2019 Article IV Staff Report) had 2021 GDP growth at 2.8 percent. Headline inflation (period-average CPI) rose to an estimated 4.5 percent in 2021, reflecting spikes in food and medical prices and the depreciation of the Kina. Partly resulting from the Porgera mine closure, the 2021 current account surplus is estimated to have

decreased to 19 percent of GDP in 2021. Over the medium-term, the forecast includes a marginally smaller surplus, around 18 percent of GDP.

Text Table 2. Papua New Guinea: DSA Vintages: Macroeconomic and Fiscal Assumptions, 2021-2032

	DSA Vintage	2020	2021	2022	2023	2024-2032 (average)
Real GDP growth y/y (in percent)	**2022**	**-3.5**	**1.2**	**4.2**	**4.7**	**3.0**
	2021	-3.5	1.7	4.8	4.3	3.1
Resource sector	2022	-8.3	-3.5	4.8	5.9	0.1
	2021	-8.4	-3.2	8.5	4.0	-0.1
Non-resource sector	2022	-1.2	3.2	4.0	4.3	4.1
	2021	-1.2	3.9	3.3	4.5	4.2
Inflation, annual average (consumer prices, percent)	**2022**	**4.9**	**4.5**	**6.4**	**5.4**	**4.6**
	2021	4.9	5.0	5.6	4.7	4.5
Current account balance (percent of GDP)	**2022**	**22.4**	**18.7**	**23.1**	**22.0**	**17.7**
	2021	21.2	20.2	22.6	21.0	21.2
Growth of exports of G&S (US$, in percent)	2022	-10.8	-3.2	26.6	3.4	2.3
	2021	-20.4	11.0	15.1	4.7	2.7
Growth of imports of G&S (US$, in percent)	2022	-22.3	-3.8	9.9	6.3	3.7
	2021	-20.7	2.2	8.7	9.4	3.3
Primary balance (percent of GDP)	**2022**	**-6.0**	**-4.3**	**-3.4**	**-2.5**	**1.4**
	2021	-6.0	-5.1	-3.7	-2.7	1.4
Government revenues (excluding grants, percent of GDP)	2022	12.5	12.4	13.1	13.3	15.0
	2021	12.5	12.7	13.8	14.0	14.6

7. Fiscal assumptions for 2022, under the SMP, include a gradual consolidation while ensuring space for needed spending is preserved. Parliament approved a budget that aims for an overall deficit of 5.9 percent of GDP, below the 6 percent target agreed with the Fund as part of the Staff Monitored Program. This budget envisages an improvement in domestic revenues, driven by a cyclical improvement in tax revenues and policy measures taken to increase the contribution from SOEs following the passage of the *Non-Tax Revenue Bill* while expenditures will be focused on supporting the recovery, continuing the shift toward capital spending, adequately financing education, and the one-off allocation for the organization of the elections. The baseline projection assumes vaccination rates remain slow, but low mobility prevents large-scale transmission between urban and rural communities. In March 2022, the authorities announced a temporary relief package to address the impacts of rising food and fuel prices triggered by the war in Ukraine consisting of tax relief and higher spending, offset by higher revenue from the mining sector due to the increase in commodity prices. As a result, the government does not envisage an increase in the fiscal deficit beyond the budget target for the year.

8. The medium-term baseline macroeconomic forecast is broadly unchanged from the 2021 DSA (Text Table 2). At 3.1 percent, the long-term potential real growth estimate remains unchanged from the 2021 DSA. Inflation is projected to remain significantly higher in the medium term, before falling to an annual average of 4.5 percent, owing largely to imported inflation from global markets. As strong global demand for PNG's export goods is expected to persist over the medium-term, and with resource-sector projects that have been delayed by the pandemic coming on stream, the current account surplus is forecast to remain very large and around 18 percent of GDP.

Text Table 3. Papua New Guinea: Summary of Fiscal Operations, 2020-2024
(In percent of GDP)

	2020	2021		2022			2023	2024
		Prog.1/	Prel.	Prog.1/	Budget 2/	Revised 3/	Proj.	
Revenue and Grants	**14.2**	**14.4**	**14.7**	**15.6**	**14.8**	**15.0**	**14.8**	**15.3**
Taxes	11.5	11.7	11.8	12.3	11.5	11.1	11.8	12.4
Grants	1.7	1.8	2.2	1.8	1.7	1.7	1.5	1.5
Other Revenue	1.0	1.0	0.7	1.5	1.7	2.2	1.5	1.4
Resource revenue 4/	0.9	0.9	1.1	1.2	1.5	2.0	1.0	1.3
Non-resource revenue	13.3	13.5	13.6	14.4	13.4	13.0	13.7	14.0
Expenditure	**22.7**	**21.8**	**21.3**	**21.6**	**20.3**	**20.5**	**19.3**	**19.0**
Expense	18.6	17.6	17.4	16.6	15.5	15.7	14.8	14.3
Compensation of employees	6.8	6.4	6.4	6.0	5.5	5.5	5.4	5.1
Use of goods and services	6.3	6.5	6.5	6.0	5.6	5.8	5.3	5.0
Interest	2.5	2.2	2.4	2.3	2.1	2.1	2.1	2.6
Other	2.9	2.5	2.1	2.3	2.3	2.3	2.1	1.6
Net acquisition of non-financial assets	4.1	4.2	3.9	5.0	4.8	4.8	4.5	4.2
Gross operating balance	-4.4	-3.2	-2.8	-1.1	-0.7	-0.7	0.0	0.5
Net lending (+)/borrowing (-)	**-8.6**	**-7.4**	**-6.6**	**-6.0**	**-5.5**	**-5.5**	**-4.5**	**-3.7**
Primary balance	-6.0	-5.1	-4.3	-3.7	-3.4	-3.4	-2.5	-1.1
Non-resource net lending (+)/borrowing (-)	-9.4	-8.3	-7.7	-7.2	-6.9	-7.5	-5.5	-5.0
Non-resource primary balance	-6.9	-6.1	-5.3	-4.9	-4.8	-5.4	-3.5	-2.4

Sources: Department of Treasury; and IMF staff estimates and projections.
Prel. = Preliminary, Proj. = IMF staff projection.
1/ Estimates/projection at the time of the approval of the Staff Monitored Progam in December 2021.
2/ Budget approved by Parliament for FY 2022 in December 2021
3/ Revised fiscal frame with fiscal relief package approved in March 2022 to addess rising food and fuel prices.
4/ Does not include projected revenues from new mining projects where the investment decision has not yet been reached.

9. The medium-term baseline also assumes rapid progress on fiscal consolidation, as envisaged by the authorities. This is appropriate to strengthen debt sustainability. While continued primary deficits are anticipated in the short-term (Text Table 3), the projection builds in a significant amount of fiscal consolidation and primary surpluses in the second half of the projection period, consistent with the authorities plans to meet the requirements of the Fiscal Responsibility Act, which specifies government debt should be maintained at no more than 40 percent of GDP over the long term. The authorities project a sharp increase in tax revenues from the PNG LNG project from 2026 onward, as tax exemptions expire. Further, the authorities see scope for significantly higher dividends after loan amortization for the project is completed but this is not yet included in the baseline. Achieving a balanced budget by 2027 would lower the risks from significant debt service obligations on external borrowing coming due in 2028.

10. The main downside risks include: outbreaks of Covid-19, natural disasters, lower global growth, and social or political instability. As new Covid-19 variants emerge, in the absence of faster progress on vaccinations, further restrictions and lockdowns are possible, with the attendant impact on growth and external inflows, particularly if the resource sector is forced to shut down again due to a high number of infections. PNG is also vulnerable to natural disasters (flooding, landslides and earthquakes) as well as the impact of climate change (through droughts and sea level rises). Lower global growth would likely impact PNG through lower commodity prices, with adverse consequences for the balance of payments and budget through lower resource revenue. Political and social risks are also elevated in the lead-up to, as well as after, the 2022 general elections. With limited sources of financing available in an adverse scenario, and continued pressing social and development needs, the room for significant policy

adjustment is relatively limited. The current SMP provided an important anchor for the authorities to advance their reform agenda, but this expires in 2022. If growth deteriorated significantly compared to the projections, further debt buildup may be needed to finance the budget and maintain government services.

11. The war in Ukraine is impacting PNG through higher commodity prices and higher inflation. This has led to a stronger balance of payments and higher fiscal revenues, given that PNG is a large commodity producer. Sustained higher commodity prices represent an upside risk to inflation and the fiscal projection. Other upside risks include the start of any of several resource projects, including Papua LNG, P'nyang LNG and the Wafi Golpu mining project, where the investment decisions have not yet been reached and thus are not included in the baseline. These could boost growth and fiscal revenues over the medium term.

12. The realism tools indicate that the government's primary balance adjustment is ambitious. At about 3.8 precent, the three-year cumulative adjustment the primary balance is within the top 12 percent of historical experiences, relative to peers. Much of this adjustment arises during 2022 and reflects temporarily higher deficits, because of the pandemic (Figure 4). Similarly, one-off factors also explain the divergence of GDP growth from the implied path consistent with the range fiscal multipliers. Normal operations are assumed to resume at the Ok-Tedi and Simberi mines in 2022, while the Porgera mine is expected to reopen in Q4 2022, contributing to strong growth in the resource sector.

13. Financing mix: For domestic financing, the DSA assumes that the composition of T-bills and T-bonds remains unchanged compared to the past six years. For the near-term, the DSA considers existing commitments and, from 2025, the DSA assumes that, on average, one-third of the financing needs are covered from official multilateral and bilateral resources, while the rest is domestic financing, with a balanced mix of bonds and T-bills. Continued development of the domestic debt market is necessary to increase liquidity and transition towards greater reliance upon domestic financing sources. Phase 2 of the Central Banking Act review is expected to look at competition in the domestic banking sector. During the SMP, there was a zero limit to new non-concessional external borrowing (NCB) by the Department of Treasury and Bank of Papua New Guinea; multilateral and official bilateral financial support is exempted from this limit. NCB may be allowed only in exceptional circumstances, which may be warranted when financing is needed for a project integral to the authorities' development program for which concessional financing is not available, or when non-concessional borrowing is used for debt management operations that improve the overall public debt profile.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TEST

Text Table 4. Papua New Guinea: Composite Indicator and Applicable Thresholds

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.871	1.11	42%
Real growth rate (in percent)	2.719	2.407	0.07	3%
Import coverage of reserves (in percent)	4.052	52.582	2.13	81%
Import coverage of reserves^2 (in percent)	-3.990	27.649	-1.10	-42%
Remittances (in percent)	2.022	0.315	0.01	0%
World economic growth (in percent)	13.520	3.050	0.41	16%
CI Score			2.62	100%
CI rating			Weak	

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Weak	Weak	Weak	Weak
	2.62	2.63	2.64

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	140
GDP	30
Debt service in % of	
Exports	10
Revenue	14

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	35

14. PNG's debt carrying capacity is assessed as weak. According to the April 2022 World Economic Outlook and Country Policy and Institutional Assessment (CPIA), PNG's Composite Indicator (CI) is 2.62, indicating weak debt-carrying capacity (Text Table 4).[3] Hence, the applicable thresholds are 30 percent for the present value (PV) of external debt-to-GDP ratio; 140 percent for the PV of the external debt-to-exports

[3] At 2.62 PNG's CI is close to the weak/medium threshold of 2.69.

ratio; 10 percent for the external debt service-to-exports ratio; 14 percent for the external debt service-to-revenue ratio; and 35 percent for the PV of public debt-to GDP ratio, respectively.

15. Scenario stress tests: As indicated in the section on public debt coverage, a contingent liabilities stress test is included to account for SOE debt not captured in official public debt data.[4] Further, given the size and importance of PNG's resource sector (with a share of commodities in total exports of goods and services of 96 percent), a commodity price shock is included in the DSA. Considering the high price volatility over the past two year, the fuel price shock is set at 35 percent (compared to the default shock of 27 percent), and the shock to non-fuel commodity prices is set to 21 percent—with 20 percent for base metals and precious metals, and 22 percent for agricultural commodities other than grain (price shocks to grain like wheat, corn, and soybeans, are not relevant for PNG and, therefore, not included in the stress test). Mitigating factors are included as well, and at default values (2 percent for fuel, and 27 percent for non-fuel). PNG's single outstanding Eurobond (maturing in 2028) activates the market financing module.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

16. Under the baseline scenario, the debt-service to revenue indicator is projected to breach its threshold. This, relatively large, breach in 2028 arises due to the bullet payment for the US$ 500 million Eurobond, which was issued in 2018. After 2028, the indicator remains close to the threshold with further marginal breaches until 2032. In the baseline scenario, the present value of the debt-to-GDP ratio as well as the debt-to-exports and debt service-to-export ratios remain below their respective thresholds over the entire projection horizon. Each of these indicators are on a downwards trend.

17. Stress tests point to vulnerabilities in PNG's external debt dynamics particularly with respect to exports shocks, which would cause threshold breaches for all four external sustainability indicators. Changes in policy and the structure of the economy manifests in a divergence between the historical scenario and the baseline. The historical scenario reflects large current account deficits associated with the construction phase of PNG LNG and is not considered an appropriate indicator for future risks. The market financing risk module indicates a moderate risk of heightened liquidity pressures as the EMBI spread is elevated, although GFNs remain below their benchmark. However, a heightened market stress event would not have a substantial impact of debt burden indicators (Figure 5). PNG's elevated sovereign spreads likely reflect the perceived risks due to the country's characteristics (small and undiversified export base, small revenue base, vulnerability to shocks).

18. As in the past, the assessment of debt dynamics is hampered by large residuals from external financial flows from money transfer by resource companies via offshore accounts (Figure

[4] In the context of the previous SMP, which expired in end-June 2021, the authorities initiated an SOE reform program to reduce the backlog of audited annual financial statements and to strengthen SOE oversight and improve understanding of fiscal risks. This program includes a detailed review of SOE debt and government guarantees and is also expected to improve the reporting of public debt.

3). The published data on the current account surplus in PNG is likely overstated due to long-standing challenges in classifying large income account outflows, including external debt service payments related to resource projects, under the financial account rather than the current account.

B. Public Sector Debt Sustainability Analysis

19. Public debt ratios have increased substantially in recent years and reached 51 percent of GDP in 2021. Starting from this level means that the public debt sustainability indicator is in breach of the threshold for countries with weak debt-carrying capacity (that is, 35 percent of GDP) during the first half of the projection horizon. However, the debt-to-revenue ratio sees a continuous downward trend over the projection horizon. In the baseline projection, the continuous fall in the public debt-to-GDP ratio over the next 5 years arises through stronger real GDP growth and a smaller fiscal deficit than over the past 5 years, as growth headwinds wane and gradual fiscal consolidation continues (Figure 3).

20. A tailored stress test for the combined contingent liability shock causes a deterioration in public debt sustainability (Figure 2 and Table 4). The trajectory of the PV of the public debt-to-GDP ratio shifts upwards by 16 percentage points from the baseline, representing the most extreme shock for all three public debt indicators. This analysis suggests contingent liabilities represent one important source of vulnerabilities in PNG.

RISK RATING AND VULNERABILITIES

21. PNG remains at "high" risk of external and overall debt distress. The (mechanical) external debt distress rating as well as the overall debt distress rating are "high", owing to the breaches of sustainability thresholds under the baseline scenario, as discussed in the previous section. No staff judgement has been applied to these ratings.

22. Debt service on existing loans, paired with relatively weak revenue generation, are expected to almost double the debt service-to-revenue ratio by 2025. However, as debt service reduces and revenues increase, and barring further shocks to demand growth, the indicator enters a significant downward trend from this peak.

23. Stress tests show that adverse shocks to exports and contingent liabilities constitute the main risks to public debt sustainability. Further, the historical scenario indicates that it will be challenging to reduce debt from current levels and that reforms, including those already implemented during the SMP, are essential for supporting the sustainability of public finances. Market financing risks continue to be relevant, with the EMBI spread threshold breached, pointing to moderate market financing pressures.

24. Debt dynamics are assessed as sustainable. Public debt is expected not to increase in the near-term and to enter a clear downward path over the medium-term. Also, the projected temporary breaches of sustainability indicators can be prevented by debt management operations as well as by boosting revenue generation. The external debt-to-GDP and debt-to-exports ratios are below their thresholds over the entire projection horizon. Public external and overall debt is judged to be sustainable conditional on the

implementation of the authorities plans for further fiscal consolidation and conservative financing strategies.

AUTHORITIES' VIEWS

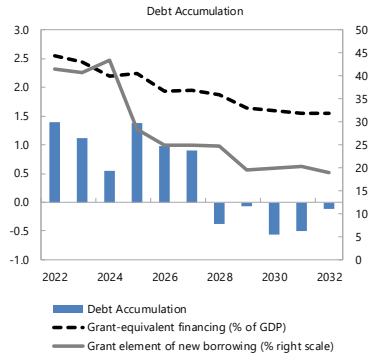
25. The authorities noted the Staff's assessment that PNG remains at high risk of debt distress but remains sustainable under the baseline projection. In discussions, the authorities agreed that large financing requirements to dampen the effects of the COVID-19 pandemic on PNG's economy have left their mark on public debt, and that redemption of the US$ 500 million Eurobond in 2028, issued in 2018, is a key risk. However, the authorities were more optimistic about their debt-carrying capacity and perceived a lower risk of debt distress, pointing to increased revenues by 2027 as debt payments for the PNG LNG project are completed. They highlighted their strategy in recent years to substitute costly financing with concessional financing from multilateral and bilateral partners, which has improved PNG's debt profile. The authorities noted interest costs of domestic securities have fallen over the past two years. They also highlighted the importance of more favorable future contract negotiation and the medium-term revenue strategy as mitigating factors, which will reduce risks in the medium-term. The authorities are committed to fiscal consolidation and conservative financing strategies to support the sustainability of PNG's debt going forward.

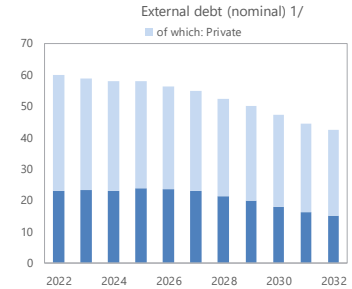
Table 1. Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2019-2042

(In percent of GDP, unless otherwise indicated)

	Actual			Projections												Average 8/	
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2042	Historical	Projections
External debt (nominal) 1/	**62.6**	**68.6**	**66.7**	**60.0**	**58.8**	**58.0**	**57.9**	**56.2**	**54.7**	**52.4**	**49.9**	**47.1**	**44.5**	**42.3**	**24.4**	**74.9**	**52.9**
of which: public and publicly guaranteed (PPG)	**18.0**	**23.0**	**24.7**	**22.9**	**23.2**	**23.1**	**23.8**	**23.5**	**23.0**	**21.3**	**19.8**	**17.9**	**16.2**	**14.9**	**6.0**	**9.2**	**20.9**
Change in external debt	-8.9	6.0	-1.9	-6.7	-1.2	-0.8	-0.1	-1.6	-1.5	-2.3	-2.5	-2.8	-2.7	-2.1	-1.8		
Identified net debt-creating flows	**-20.0**	**-22.5**	**-25.7**	**-25.8**	**-24.9**	**-22.6**	**-21.8**	**-20.9**	**-20.0**	**-19.5**	**-18.6**	**-17.8**	**-17.1**	**-16.3**	**-10.4**	**-13.2**	**-20.5**
Non-interest current account deficit	**-20.4**	**-24.2**	**-20.1**	**-24.4**	**-23.1**	**-21.6**	**-20.7**	**-19.7**	**-20.0**	**-19.5**	**-18.6**	**-17.8**	**-17.0**	**-16.3**	**-10.3**	**-13.1**	**-19.9**
Deficit in balance of goods and services	-22.7	-23.0	-20.2	-24.3	-23.3	-21.8	-21.0	-20.0	-19.2	-18.6	-17.9	-17.2	-16.4	-15.7	-10.1	**-13.1**	**-19.6**
Exports	45.8	41.0	35.8	39.2	38.2	36.8	35.7	34.5	33.2	32.5	31.4	30.4	29.3	28.3	20.1		
Imports	23.1	18.0	15.7	14.9	14.9	15.1	14.8	14.4	14.0	13.9	13.5	13.2	12.9	12.6	10.0		
Net current transfers (negative = inflow)	-0.3	-1.2	-1.0	-1.3	-1.2	-1.4	-1.3	-1.2	-1.2	-1.1	-1.0	-1.0	-0.9	-0.9	-0.5	**-0.9**	**-1.1**
of which: official	-1.4	-1.2	-1.1	-1.4	-1.3	-1.3	-1.2	-1.1	-1.1	-1.0	-1.0	-0.9	-0.9	-0.8	-0.4		
Other current account flows (negative = net inflow)	2.6	0.1	1.1	1.3	1.4	1.5	1.5	1.5	0.3	0.3	0.3	0.3	0.3	0.3	0.2	**0.9**	**0.8**
Net FDI (negative = inflow)	**0.1**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.2**	**-0.2**	**-0.2**	**-0.2**	**-0.2**	**-0.2**	**-0.2**	**-0.2**	**-0.1**	**0.7**	**-0.2**
Endogenous debt dynamics 2/	**0.3**	**2.0**	**-5.3**	**-1.2**	**-1.5**	**-0.7**	**-0.8**	**-0.9**	**0.2**	**0.2**	**0.2**	**0.2**	**0.1**	**0.1**	**0.1**		
Contribution from nominal interest rate	2.1	1.8	1.4	1.2	1.1	1.0	0.8	0.7	1.8	1.8	1.7	1.6	1.5	1.4	0.9		
Contribution from real GDP growth	-3.1	2.2	-0.7	-2.4	-2.7	-1.7	-1.6	-1.6	-1.6	-1.6	-1.5	-1.4	-1.4	-1.3	-0.8		
Contribution from price and exchange rate changes	1.3	-2.0	-6.0	...	...	...	...	...	...	...	...	...	...	...	...		
Residual 3/	**11.2**	**28.5**	**23.8**	**19.1**	**23.7**	**21.7**	**21.7**	**19.3**	**18.5**	**17.1**	**16.2**	**15.0**	**14.4**	**14.2**	**8.5**	**12.5**	**18.3**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators																	
PV of PPG external debt-to-GDP ratio	**...**	**...**	**23.1**	**21.2**	**21.1**	**20.8**	**21.2**	**20.9**	**20.5**	**19.2**	**18.0**	**16.4**	**14.9**	**13.9**	**5.6**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**64.6**	**54.1**	**55.1**	**56.4**	**59.2**	**60.6**	**61.7**	**59.0**	**57.2**	**53.9**	**50.8**	**49.0**	**28.1**		
PPG debt service-to-exports ratio	**1.2**	**4.8**	**4.5**	**2.7**	**3.5**	**3.9**	**4.6**	**5.0**	**5.0**	**8.5**	**6.7**	**8.1**	**7.7**	**7.4**	**6.0**		
PPG debt service-to-revenue ratio	**3.8**	**15.8**	**12.8**	**7.9**	**10.2**	**10.3**	**11.5**	**11.6**	**11.1**	**18.7**	**13.9**	**15.9**	**14.4**	**13.2**	**7.6**		
Gross external financing need (Million of U.S. dollars)	-2319.0	-3906.7	-3298.9	-5617.9	-5343.7	-5404.7	-5262.8	-6134.5	-5612.5	-5254.2	-5605.6	-5573.9	-5747.5	-5949.5	-7497.7		
Key macroeconomic assumptions																	
Real GDP growth (in percent)	4.5	-3.5	1.2	4.2	4.7	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.1	3.1	3.4	**4.0**	**3.3**
GDP deflator in US dollar terms (change in percent)	-1.7	3.3	9.5	10.9	1.3	1.1	1.5	3.0	3.2	1.9	3.2	3.2	3.3	3.3	3.2	**0.7**	**3.3**
Effective interest rate (percent) 4/	3.1	2.8	2.2	2.1	2.0	1.8	1.5	1.3	3.5	3.5	3.4	3.4	3.4	3.3	3.6	**2.7**	**2.6**
Growth of exports of G&S (US dollar terms, in percent)	12.7	-10.8	-3.2	26.6	3.4	0.3	1.5	2.2	2.5	2.7	2.7	2.8	2.9	2.9	3.2	**4.4**	**4.6**
Growth of imports of G&S (US dollar terms, in percent)	43.3	-22.3	-3.8	9.9	6.3	5.1	2.5	3.4	3.6	3.6	3.7	3.7	3.9	4.0	4.3	**-4.0**	**4.5**
Grant element of new public sector borrowing (in percent)	...	...	...	41.5	40.7	43.3	28.4	24.9	24.9	24.8	19.6	19.9	20.2	19.0	19.6	**...**	**27.9**
Government revenues (excluding grants, in percent of GDP)	14.2	12.5	12.4	13.1	13.3	13.8	14.4	14.7	14.8	14.8	15.1	15.5	15.8	16.0	16.0	**15.7**	**14.7**
Aid flows (in Million of US dollars) 5/	1258.7	1736.2	1119.8	917.6	975.9	816.9	814.3	764.8	813.6	808.6	697.6	719.8	742.7	766.4	1006.6		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.6	2.4	2.2	2.2	1.9	2.0	1.9	1.6	1.6	1.5	1.5	1.1	**...**	**2.0**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	67.2	63.7	70.0	52.8	54.5	56.7	57.1	59.5	59.9	60.2	54.5	59.9	**...**	**59.6**
Nominal GDP (Million of US dollars)	24,751	24,670	27,344	31,603	33,524	34,909	36,486	38,701	41,139	43,181	45,898	48,800	51,977	55,402	105,634		
Nominal dollar GDP growth	2.7	-0.3	10.8	15.6	6.1	4.1	4.5	6.1	6.3	5.0	6.3	6.3	6.5	6.6	6.7	**4.5**	**6.7**
Memorandum items:																	
PV of external debt 7/	...	...	65.1	58.3	56.7	55.7	55.2	53.6	52.2	50.3	48.1	45.6	43.2	41.3	24.1		
In percent of exports	...	...	181.8	148.6	148.3	151.2	154.5	155.7	157.3	154.6	153.2	150.2	147.3	145.7	119.9		
Total external debt service-to-exports ratio	23.9	21.0	23.2	17.5	19.4	17.4	18.4	12.0	19.9	23.2	21.0	21.7	21.0	20.1	16.5		
PV of PPG external debt (in Million of US dollars)			6326.8	6707.7	7058.5	7244.1	7727.2	8085.1	8432.9	8278.0	8245.8	7987.3	7741.5	7682.0	5955.3		
(PVt-PVt-1)/GDPt-1 (in percent)				1.4	1.1	0.6	1.4	1.0	0.9	-0.4	-0.1	-0.6	-0.5	-0.1	-0.2		
Non-interest current account deficit that stabilizes debt ratio	-11.6	-30.2	-18.2	-17.7	-21.9	-20.8	-20.6	-18.1	-18.6	-17.2	-16.1	-15.0	-14.4	-14.1	-8.5		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

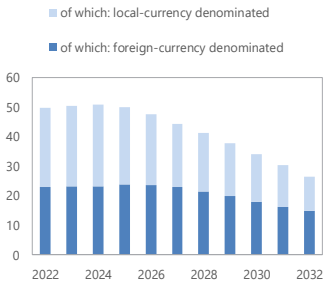
Table 2. Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2019-2042

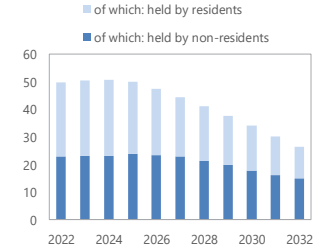
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
Public sector debt 1/	**41.5**	**49.0**	**51.4**	**49.8**	**50.3**	**50.7**	**49.8**	**47.4**	**44.3**	**26.3**	**4.5**	**31.5**	**42.0**
of which: external debt	**18.0**	**23.0**	**24.7**	**22.9**	**23.2**	**23.1**	**23.8**	**23.5**	**23.0**	**14.9**	**6.0**	**9.2**	**20.9**
Change in public sector debt	4.7	7.5	2.4	-1.6	0.6	0.4	-0.9	-2.4	-3.1	-3.9	-1.5		
Identified debt-creating flows	**2.3**	**2.4**	**-2.6**	**-0.7**	**0.2**	**0.1**	**-0.9**	**-2.1**	**-2.7**	**-3.6**	**-2.2**	**0.5**	**-2.0**
Primary deficit	**1.9**	**6.0**	**4.3**	**3.4**	**2.5**	**1.1**	**0.0**	**-0.9**	**-1.6**	**-2.7**	**-1.8**	**2.9**	**-0.6**
Revenue and grants	16.3	14.2	14.7	14.8	14.8	15.3	15.8	16.1	16.3	17.2	16.9	**17.6**	**16.1**
of which: grants	2.1	1.7	2.2	1.7	1.5	1.5	1.4	1.4	1.5	1.2	0.9		
Primary (noninterest) expenditure	18.2	20.2	18.9	18.2	17.2	16.4	15.8	15.2	14.7	14.5	15.1	**20.5**	**15.4**
Automatic debt dynamics	**0.4**	**2.4**	**-2.6**	**-4.1**	**-2.3**	**-1.0**	**-0.9**	**-1.2**	**-1.1**	**-0.9**	**-0.4**		
Contribution from interest rate/growth differential	0.2	2.6	-1.6	-4.1	-2.3	-1.0	-0.9	-1.2	-1.1	-0.9	-0.4		
of which: contribution from average real interest rate	1.7	1.1	-1.0	-2.0	0.0	0.4	0.6	0.3	0.3	0.0	-0.2		
of which: contribution from real GDP growth	-1.6	1.5	-0.6	-2.1	-2.3	-1.4	-1.5	-1.4	-1.4	-0.9	-0.2		
Contribution from real exchange rate depreciation	0.3	-0.2	-1.1	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**2.4**	**5.1**	**5.0**	**2.4**	**2.8**	**1.4**	**0.0**	**-1.3**	**-0.4**	**-0.3**	**0.7**	**3.7**	**0.3**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**50.2**	**48.4**	**48.5**	**48.7**	**47.5**	**45.1**	**42.1**	**25.3**	**4.2**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**342.4**	**326.6**	**328.3**	**317.8**	**300.6**	**279.6**	**258.3**	**146.6**	**24.9**		
Debt service-to-revenue and grants ratio 3/	**106.8**	**131.0**	**48.4**	**35.4**	**50.8**	**67.7**	**78.1**	**75.8**	**70.8**	**40.3**	**-10.4**		
Gross financing need 4/	19.3	24.6	11.3	8.6	10.0	11.5	12.3	11.3	9.9	4.2	-3.6		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.5	-3.5	1.2	4.2	4.7	3.0	3.0	3.0	3.0	3.1	3.4	**4.0**	**3.3**
Average nominal interest rate on external debt (in percent)	0.4	2.7	1.7	2.2	2.6	2.6	2.6	2.6	2.6	2.0	2.2	**1.4**	**2.4**
Average real interest rate on domestic debt (in percent)	9.4	3.5	-1.7	-4.1	0.3	1.5	1.7	0.6	0.6	-0.1	0.1	**3.9**	**0.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	1.9	-1.2	-4.9	...	...	...	...	...	...	...	...	**2.9**	**...**
Inflation rate (GDP deflator, in percent)	1.1	5.5	9.5	10.9	1.3	1.1	1.5	3.0	3.2	3.3	3.2	**4.3**	**3.4**
Growth of real primary spending (deflated by GDP deflator, in percent)	5.8	7.0	-5.3	0.2	-0.7	-1.7	-1.1	-0.6	-0.9	3.1	3.9	**3.0**	**0.8**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-2.9	-1.5	1.9	5.0	1.9	0.7	0.9	1.5	1.5	1.2	-0.3	**-0.8**	**1.6**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

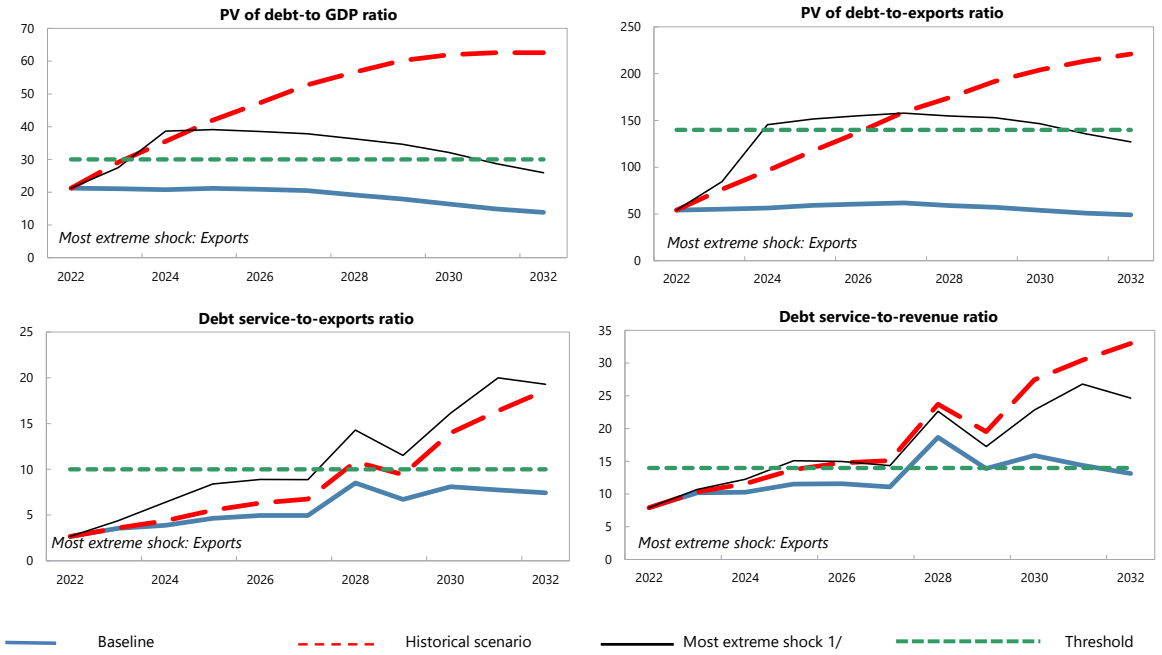
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2022-2032



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price	Yes	No
Market financing	No	No

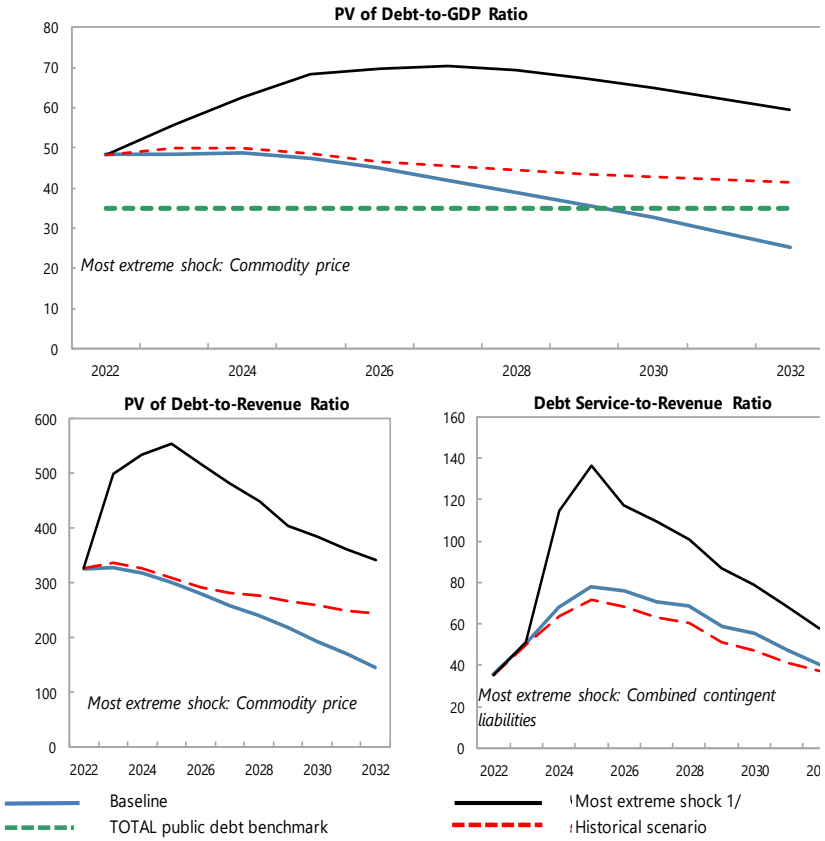
Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	16	16
Avg. grace period	6	6

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2022-2032



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	22%	22%
Domestic medium and long-term	37%	37%
Domestic short-term	40%	40%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
Avg. maturity (incl. grace period)	16	16
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.0%	4.0%
Avg. maturity (incl. grace period)	2	2
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	2.2%	2.2%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2022-2032

(In percent)

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of debt-to GDP ratio											
Baseline	21	21	21	21	21	20	19	18	16	15	14
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	21	29	**36**	**42**	**47**	**53**	**57**	**60**	**62**	**63**	**63**
B. Bound Tests											
B1. Real GDP growth	21	23	24	24	24	24	22	21	19	17	16
B2. Primary balance	21	21	22	23	23	23	22	21	19	18	17
B3. Exports	21	28	**39**	**39**	**39**	**38**	**36**	**35**	**32**	29	26
B4. Other flows 3/	21	24	27	27	27	27	25	24	22	20	18
B5. Depreciation	21	27	28	28	28	28	26	24	22	20	19
B6. Combination of B1-B5	21	28	**31**	**31**	**31**	**30**	29	27	25	22	21
C. Tailored Tests											
C1. Combined contingent liabilities	21	24	25	26	27	27	27	26	25	24	23
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	21	27	**32**	**34**	**36**	**37**	**37**	**37**	**36**	**35**	**33**
C4. Market Financing	21	23	23	24	23	23	21	20	18	17	15
Threshold	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio											
Baseline	54	55	56	59	61	62	59	57	54	51	49
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	54	76	96	117	137	**159**	**174**	**192**	**204**	**214**	**221**
B. Bound Tests											
B1. Real GDP growth	54	55	56	59	61	62	59	57	54	51	49
B2. Primary balance	54	56	60	64	67	69	67	66	63	60	59
B3. Exports	54	84	**146**	**152**	**155**	**158**	**155**	**153**	**147**	136	127
B4. Other flows 3/	54	63	73	77	78	80	77	76	72	67	64
B5. Depreciation	54	55	60	63	65	66	63	61	58	55	52
B6. Combination of B1-B5	54	74	75	96	98	100	97	95	90	84	79
C. Tailored Tests											
C1. Combined contingent liabilities	54	62	67	74	78	82	82	83	82	80	80
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	54	82	100	108	114	119	119	122	123	122	121
C4. Market Financing	54	55	56	59	61	62	59	57	54	51	49
Threshold	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio											
Baseline	3	4	4	5	5	5	9	7	8	8	7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	3	4	4	6	6	7	**11**	9	**14**	**16**	**19**
B. Bound Tests											
B1. Real GDP growth	3	4	4	5	5	5	9	7	8	8	7
B2. Primary balance	3	4	4	5	5	5	9	7	8	8	8
B3. Exports	3	4	6	8	9	9	**14**	**12**	**16**	**20**	**19**
B4. Other flows 3/	3	4	4	5	5	5	9	7	9	**10**	10
B5. Depreciation	3	4	4	5	5	5	9	7	8	8	8
B6. Combination of B1-B5	3	4	5	6	7	7	**11**	9	**12**	**13**	**12**
C. Tailored Tests											
C1. Combined contingent liabilities	3	4	4	5	5	5	9	7	9	8	8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	3	4	5	6	7	7	**11**	9	**12**	**14**	**14**
C4. Market Financing	3	4	4	5	5	5	9	7	8	7	7
Threshold	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio											
Baseline	8	10	10	12	12	11	**19**	14	**16**	**14**	13
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	8	10	12	14	**15**	**15**	**24**	**20**	**27**	**30**	**33**
B. Bound Tests											
B1. Real GDP growth	8	11	12	13	13	13	**21**	**16**	**18**	**17**	**15**
B2. Primary balance	8	10	10	12	12	12	**19**	**14**	**17**	**16**	**14**
B3. Exports	8	11	12	**15**	**15**	**14**	**23**	**17**	**23**	**27**	**25**
B4. Other flows 3/	8	10	11	12	12	12	**19**	**15**	**18**	**19**	**17**
B5. Depreciation	8	13	13	**15**	**15**	**14**	**24**	**18**	**20**	**19**	**18**
B6. Combination of B1-B5	8	11	12	**14**	**14**	13	**22**	**17**	**22**	**21**	**20**
C. Tailored Tests											
C1. Combined contingent liabilities	8	10	11	12	12	12	**20**	**15**	**17**	**15**	**14**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	8	**14**	**16**	**19**	**18**	**16**	**24**	**18**	**23**	**24**	**23**
Threshold	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

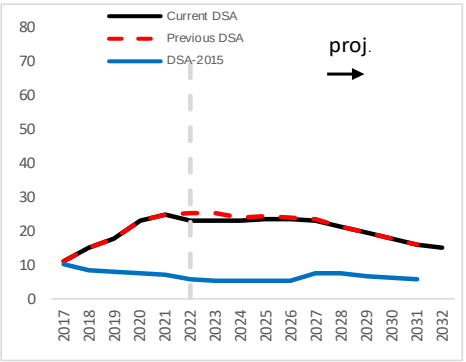
Table 4. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt, 2022-2032

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of Debt-to-GDP Ratio											
Baseline	**48**	**48**	**49**	**48**	**45**	**42**	**39**	**36**	33	29	25
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	**48**	**50**	**50**	**49**	**47**	**45**	**44**	**44**	**43**	**42**	**41**
B. Bound Tests											
B1. Real GDP growth	**48**	**56**	**63**	**63**	**61**	**59**	**58**	**56**	**54**	**51**	**48**
B2. Primary balance	**48**	**53**	**58**	**56**	**52**	**49**	**46**	**42**	**39**	**35**	31
B3. Exports	**48**	**54**	**63**	**62**	**59**	**56**	**53**	**49**	**45**	**40**	35
B4. Other flows 3/	**48**	**52**	**55**	**54**	**51**	**48**	**45**	**42**	**38**	34	29
B5. Depreciation	**48**	**54**	**54**	**52**	**47**	**43**	**39**	**35**	31	27	22
B6. Combination of B1-B5	**48**	**51**	**55**	**54**	**50**	**47**	**44**	**41**	**38**	35	31
C. Tailored Tests											
C1. Combined contingent liabilities	**48**	**67**	**68**	**66**	**62**	**58**	**55**	**51**	**47**	**43**	**39**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**48**	**56**	**63**	**68**	**70**	**70**	**70**	**67**	**65**	**62**	**59**
C4. Market Financing	**48**	**48**	**49**	**48**	**45**	**42**	**39**	**36**	33	29	25
TOTAL public debt benchmark	35	35	35	35	35	35	35	35	35	35	35
PV of Debt-to-Revenue Ratio											
Baseline	327	328	318	301	280	258	240	217	194	170	147
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	327	338	327	310	291	282	277	267	258	249	244
B. Bound Tests											
B1. Real GDP growth	327	379	406	396	375	361	351	333	316	296	278
B2. Primary balance	327	358	376	355	325	302	282	257	232	205	181
B3. Exports	327	364	412	392	368	344	325	299	270	235	203
B4. Other flows 3/	327	349	359	341	318	296	277	253	227	198	171
B5. Depreciation	327	370	357	330	296	268	243	215	186	157	130
B6. Combination of B1-B5	327	345	356	339	312	291	273	250	227	202	178
C. Tailored Tests											
C1. Combined contingent liabilities	327	453	442	416	383	357	337	309	282	254	227
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	327	500	533	555	515	481	449	404	384	362	342
C4. Market Financing	327	328	318	301	280	259	240	217	194	170	147
Debt Service-to-Revenue Ratio											
Baseline	35	51	68	78	76	71	69	59	55	48	40
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	35	50	64	71	68	63	60	51	47	41	37
B. Bound Tests											
B1. Real GDP growth	35	55	80	98	102	101	103	96	94	89	83
B2. Primary balance	35	51	73	95	93	84	81	69	64	56	48
B3. Exports	35	51	68	80	78	73	70	61	60	57	49
B4. Other flows 3/	35	51	68	79	77	72	70	60	57	52	44
B5. Depreciation	35	49	66	76	74	69	70	59	56	48	41
B6. Combination of B1-B5	35	50	69	81	79	75	74	64	61	54	46
C. Tailored Tests											
C1. Combined contingent liabilities	35	51	115	137	117	109	101	87	79	68	58
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	35	69	93	110	118	123	122	109	107	102	98
C4. Market Financing	35	51	68	78	76	71	69	60	55	47	40

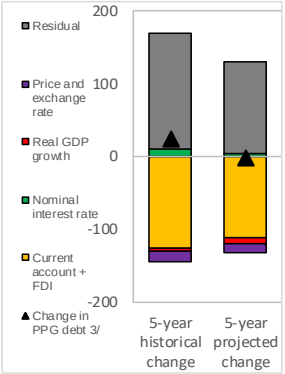
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 3. Papua New Guinea: Drivers of Debt Dynamics – Baseline Scenario

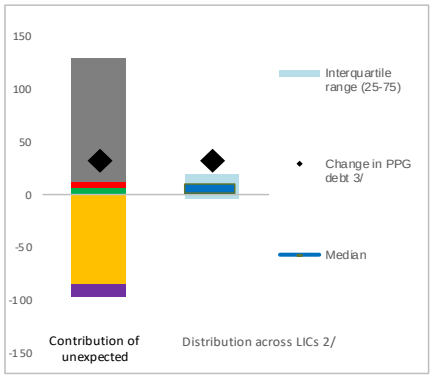
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

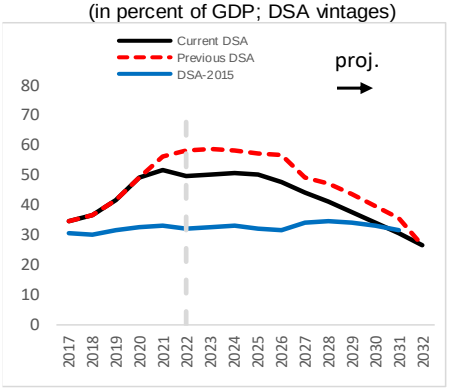


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

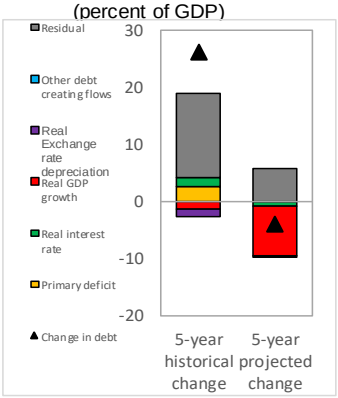


Public debt

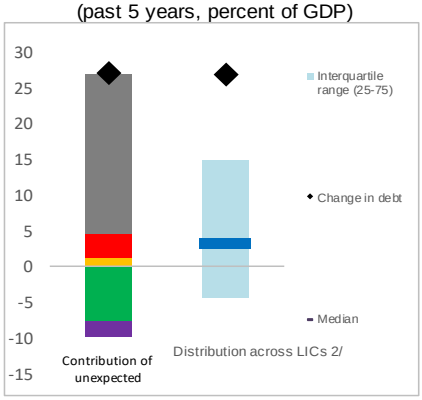
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



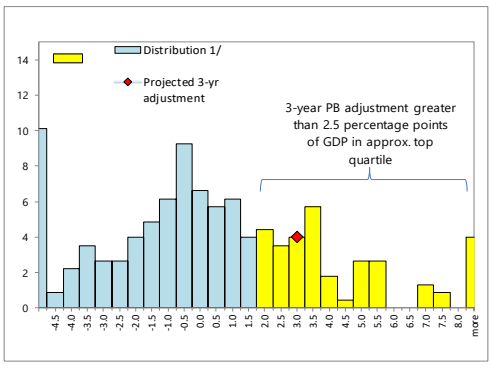
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

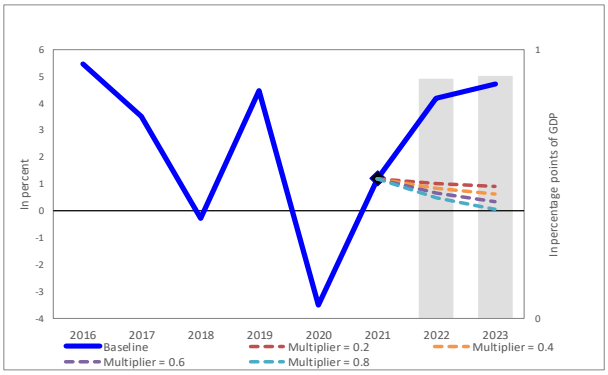
Figure 4. Papua New Guinea: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)

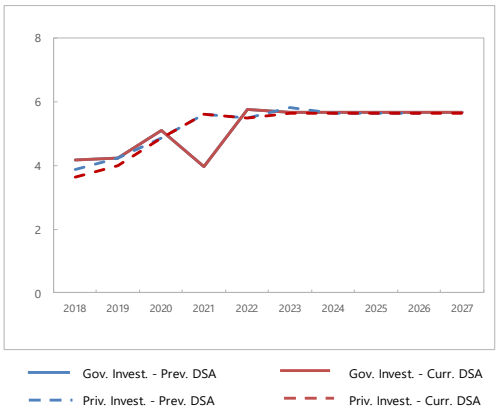
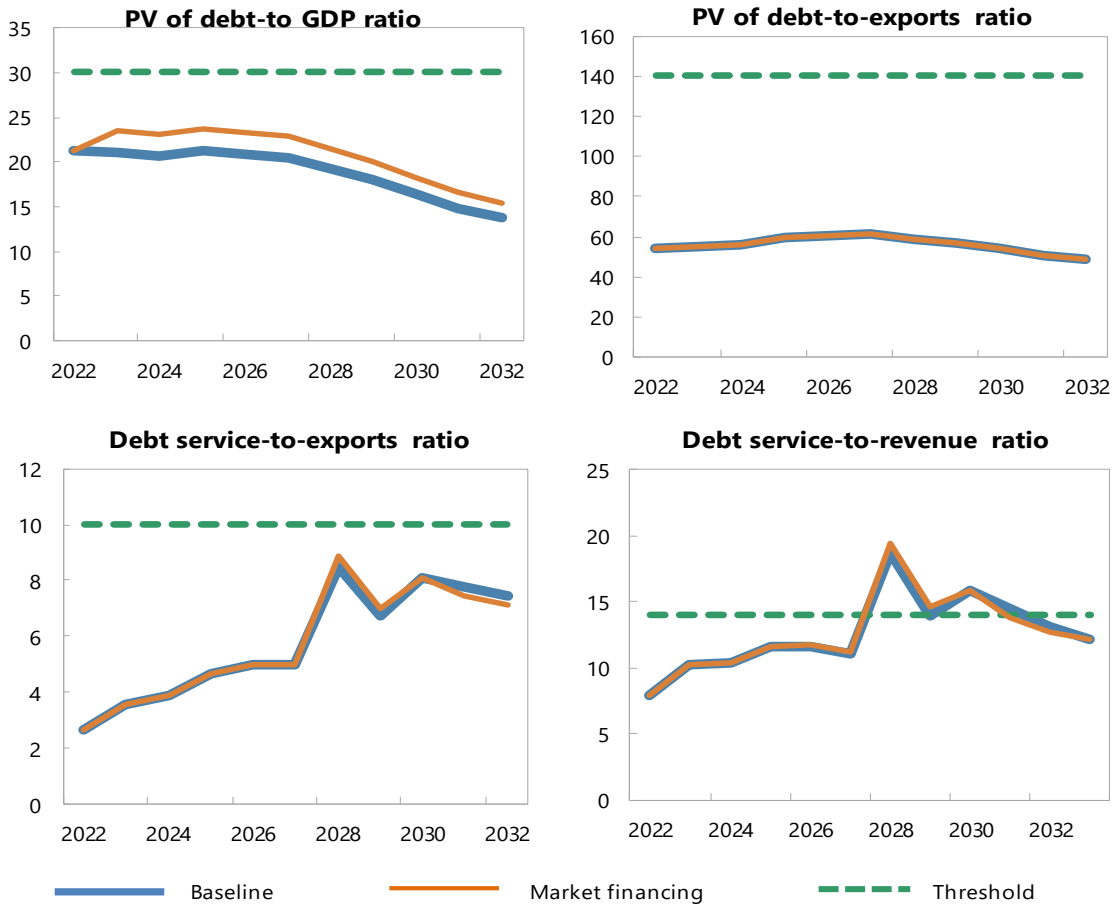


Figure 5. Papua New Guinea: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	11	854
Breach of benchmark	No	Yes
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.